

FAITH-SAE: Are Sparse-Autoencoder Concept Directions in Vision Models Causally Faithful Under Distribution Shift?

8-Week Research Roadmap

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1 What This Project Is About

Project Snapshot

Building blocks	Steering methods over a <i>frozen</i> CLIP ViT-B/16 + TopK sparse autoencoder: naive_steer (off-manifold activation-addition), random_steer , clamp_steer , and onmanifold_steer (project the edit onto the top- r real-image subspace, <i>ours</i>).
Baselines	Supervised TCAV-style concept-direction steering (quality reference), naive off-manifold activation-addition (main competitor), random-direction (null/sanity), raw-clamp — all at matched steering strength .
Benchmarks	Clean ImageNet-val → ImageNet-R → ImageNet-Sketch → ImageNet-C (severity 1–5) → ObjectNet.
Metrics	Monotonicity, specificity, sufficiency → Causal Faithfulness Score (CFS) $\in [0, 1]$; off-manifold residual; Δ CFS-per-shift degradation slope with bootstrap CIs over concepts.
Big idea	A steering <i>method</i> with no metric is unfalsifiable; a <i>metric</i> applied to naive edits just measures artifacts. Combine them: measure CFS of on-manifold steering <i>across the OOD ladder</i> . Each mechanism cures the other’s blind spot.
Target	On-manifold steering achieves higher CFS than naive / random / clamp at matched strength on clean images, approaches the supervised reference, and degrades more gracefully under shift — locating the collapse knee.

Sparse autoencoders (SAEs) decompose a frozen backbone’s dense, polysemantic activations into a large dictionary of sparse, human-nameable *concept directions*. In vision this has produced striking demonstrations — directions that appear to encode “stripes,” “dog face,” or “snowy scene.” But a picture that *looks* interpretable is not the same as a direction that is *causally real*: if we push the “dog face” direction up, does behaviour move smoothly and specifically the way the concept’s name promises, or have we merely produced a plausible artifact?

This project studies two ingredients usually examined apart. **Mechanism A — on-manifold steering (the method)** forces a chosen SAE concept up or down inside the frozen backbone, but *constrains* the edit: instead of the raw activation-addition Δ , it adds the projection $P_M \cdot \Delta$ onto the top- r principal directions of *real*-image activations, so the steered activation stays in the region the model was trained on. The edit is realistic, decodable, and non-destructive. Its weakness alone: a steering method with no metric is unfalsifiable. **Mechanism B — the CFS metric plus OOD stress (the measuring**

stick) is a single score in $[0, 1]$ asking whether an edit is *monotone* (knob up $\rightarrow$ readout up smoothly), *specific* (only the target concept moves), and *sufficient* (effect size matches the claimed magnitude) — re-measured as images get harder. Its weakness alone: on naive steering it just quantifies off-manifold artifacts.

The opening

The unoccupied space is the **combination**: measure the CFS of on-manifold steering *across* the OOD sweep (clean $\rightarrow$ R $\rightarrow$ Sketch $\rightarrow$ C $\rightarrow$ ObjectNet). On-manifold steering gives the metric something *real* to measure; the metric proves the on-manifold edit is faithful and tracks exactly where it breaks. The CFS-versus-shift curve is the paper’s answer. Prior work shows SAEs *find* vision concepts and that you *can* steer them, and separately warns that off-manifold steering is unreliable — but no one has a **quantitative faithfulness benchmark for vision SAE features**, and no one has tested faithfulness **under distribution shift**.

Scope & Realism Note

This is a *controlled, reproducible* interpretability study, not a leaderboard chase. We use one small, public, frozen backbone (CLIP ViT-B/16) and a fixed public data budget (ImageNet-val and its shifted variants), so every comparison holds the backbone, SAE, and steering strength constant and changes only the steering *method* or one design knob. The headline quantity — the **Causal Faithfulness Score**, a backbone- and library-independent number per concept and per shift level (the analog of “bytes/token”) — is reported alongside wall-clock so results transfer beyond our setup. Every figure ships with *illustrative* placeholder numbers until the real CLIP + ImageNet-shift runs land at Milestone 3.

2 Research Questions & Timeline

RQ1 (headline — on-manifold vs. the field). Does on-manifold projected steering achieve a higher Causal Faithfulness Score (CFS) than naive off-manifold activation-addition steering — and than random-direction, raw-clamp, and supervised concept-direction (TCAV-style) steering — at **matched steering strength** on clean, in-distribution images?

RQ2 (core knob — decomposition & design). How does CFS *decompose* into monotonicity / specificity / sufficiency, what fraction of discovered concepts steer reliably (the field’s “only $\sim 10\text{--}15\%$ ” claim), and how does the faithfulness optimum depend on the two design knobs — **steering strength** s and **manifold-projection rank** r ?

RQ3 (the sweep — collapse knee). Does CFS *survive distribution shift* across clean ImageNet $\rightarrow$ ImageNet-R $\rightarrow$ ImageNet-Sketch $\rightarrow$ ImageNet-C (corruption-severity dial) $\rightarrow$ ObjectNet, and **where is the collapse knee** — i.e. what is the ΔCFS -per-shift-level degradation slope, and does on-manifold steering degrade more gracefully than naive?

Milestone	Weeks	Key Deliverable(s)
M1 — Literature Review & Block Design	W1–2	Annotated 12-paper review; CFS definition fixed; the four steering variants and the evaluation protocol designed; Deliverable #1.
M2 — Data Pipeline & Implementation	W3–4	CLIP ViT-B/16 + TopK SAE pipeline; the OOD data ladder loaded; all variants implemented and validated on the synthetic planted-concept task; Deliverable #2.
M3 — Training & Ablation Experiments	W5–6	Full CFS measurements across variants and the OOD ladder; ablations A1–A5; error taxonomy populated; <code>results/metrics_all.csv</code> ; Deliverable #3.
M4 — Analysis, Insights & Manuscript	W7–8	9 figures; insight report answering RQ1–RQ3; practitioner decision table; IEEE-conference manuscript draft; Deliverable #4.

3 Milestone 1: Literature Review & Block Design

Objectives (three pillars).

1. **Concepts.** Master how vision SAEs discover concept directions (TopK vs. L1 dictionaries, monosemanticity, the polysemantic tail) so we know what we are steering and why most directions are not clean knobs.
2. **Steering & faithfulness.** Pin down activation-addition / representation engineering, the on-manifold constraint $P_M \cdot \Delta$, and what it means for an edit to be causally faithful (monotone, specific, sufficient) — then formalize CFS as a single conjunctive score.
3. **Robustness.** Understand the distribution-shift ladder (renditions, sketch, graded corruption, real-world ObjectNet shift) so the OOD sweep is a principled stress test, not an arbitrary sample.

Component	Mechanism	Role
TopK Sparse Autoencoder	Sparse dictionary over frozen ViT activations; keep top- k features active per token	Discovers the concept directions we steer
Concept direction d	A single SAE decoder column (a feature’s direction in activation space)	The knob we push up/down
On-manifold projection P_M	$P_M = U_r U_r^\top$ onto top- r real-image PCA subspace; $a' = a + s(P_M \Delta)$	Keeps the edit realistic / decodable (the method)
Steering strength s	Scalar multiplier on the (projected) edit	The intensity knob; too large $\Rightarrow$ saturation
CFS = HM(mono, spec, suff)	Harmonic mean $\Rightarrow$ conjunctive: faithful only if <i>all three</i> hold	The single contributed measuring stick
OOD ladder	clean $\rightarrow$ R $\rightarrow$ Sketch $\rightarrow$ C(1–5) $\rightarrow$ ObjectNet	The shift axis; yields the CFS-vs-severity curve

Reading list (12 papers)

1. Cunningham et al. 2023 — *Sparse Autoencoders Find Highly Interpretable Features* — arXiv:2309.08600.
2. Bricken et al. 2023 — *Towards Monosemanticity: Decomposing Language Models with Dictionary Learning* (Anthropic).

3. Gao et al. 2024 — *Scaling and Evaluating Sparse Autoencoders* (TopK SAE) — arXiv:2406.04093.
4. Templeton et al. 2024 — *Scaling Monosemanticity* (Anthropic).
5. Turner et al. 2023 — *Activation Addition / ActAdd: Steering Language Models* — arXiv:2308.10248.
6. Zou et al. 2023 — *Representation Engineering: A Top-Down Approach to AI Transparency* — arXiv:2310.01405.
7. Kim et al. 2017 — *Interpretability Beyond Feature Attribution (TCAV)* — arXiv:1711.11279.
8. Radford et al. 2021 — *Learning Transferable Visual Models from Natural Language (CLIP)* — arXiv:2103.00020.
9. Dosovitskiy et al. 2020 — *An Image is Worth 16x16 Words (ViT)* — arXiv:2010.11929.
10. Hendrycks et al. 2020 — *The Many Faces of Robustness (ImageNet-R)* — arXiv:2006.16241.
11. Hendrycks & Dietterich 2019 — *Benchmarking Neural Network Robustness (ImageNet-C)* — arXiv:1903.12261.
12. Wang et al. 2019 — *Learning Robust Global Representations (ImageNet-Sketch)* — arXiv:1905.13549.

The manuscript expands this to ~ 24 entries with ObjectNet (Barbu et al. 2019), Causal Scrubbing (Chan et al. 2022), Makelov et al. (2405.08366), Rajamanoharan et al. *JumpReLU / Gated SAEs* (2407.14435 / 2404.16014), Marks et al. *Sparse Feature Circuits* (2403.19647), Fel et al. (vision dictionary learning), Hooker et al. *ROAR* (1806.10758), and Geirhos et al. *Texture-vs-Shape Bias* (1811.12231).

Core Research Insight

Faithfulness is **conjunctive**, not additive. A steered concept is trustworthy only if it is *monotone and specific and sufficient at once* — a near-zero on any single axis (e.g. an edit that moves the readout but also drags every off-target probe) should collapse the whole score. We therefore define CFS as the *harmonic mean* of the three components, so one weak axis dominates. The on-manifold projection $P_M \Delta$ is what gives that conjunction a chance to hold: by keeping the edit inside the real-image subspace, it preserves specificity and decodability that naive off-manifold addition destroys — and we expect that advantage to widen, not shrink, as inputs move out of distribution.

Evaluation plan

Metrics. Per concept: monotonicity (Spearman correlation between the steering knob and a held-out concept readout), specificity (1 – normalized drift of off-target linear probes), sufficiency (standardized Cohen’s- d effect size at full knob vs. the concept’s claimed magnitude); composite CFS = HM($\cdot$); off-manifold residual $\|\Delta - P_M \Delta\| / \|\Delta\|$; and the OOD degradation slope $\Delta \text{CFS} / \Delta \text{shift}$ with bootstrap CIs over concepts. **Protocols.** Matched steering strength across all methods; held-out readout probes never seen during direction selection; the same selected concept set evaluated at every shift level. **Baselines.** Supervised TCAV-style (quality reference), naive off-manifold (main competitor), random-direction (null), raw-clamp.

Deliverable #1 — Review & Design

A written literature review of the 12 papers; a frozen, equation-level definition of CFS and the on-manifold edit $a' = a + s(P_M \Delta)$; the four steering variants (**naive_steer**, **random_steer**, **clamp_steer**, **onmanifold_steer**) and the supervised reference specified; the evaluation protocol (metrics, matched-strength rule, OOD ladder) fixed.

Acceptance Check

CFS and $P_M\Delta$ are written down unambiguously; each of the 5 baselines/variants has a one-line precise definition; the protocol guarantees every method is compared at matched strength on the *same* concepts; a reviewer can restate the open gap (no quantitative vision-SAE faithfulness benchmark; none tested under shift) in one sentence.

4 Milestone 2: Data Pipeline & Implementation

Dataset	Size	Description	Use
ImageNet-val	50k	Clean in-distribution validation images	Quality reference: clean CFS, concept readouts, SAE training source
ImageNet-R	30k	Renditions / abstractions (art, cartoons, sculpture)	Does faithfulness survive style/domain shift
ImageNet-Sketch	50k	Black-and-white sketches; texture removed	Concept readout without texture cues (hard stress)
ImageNet-C	50k×	15 corruptions × severity 1–5 (the dial)	Graded OOD: the headline CFS-vs-severity curve
ObjectNet	50k	Real-world pose / background / view-point controls	Hardest shift; the faithfulness floor

Tasks.

1. Load frozen CLIP ViT-B/16; extract & cache patch (and CLS) activations at the chosen layer for ImageNet-val.
2. Train a TopK SAE; build the concept dictionary; estimate the top- r real-image PCA subspace U_r for P_M .
3. Implement the four steering variants behind one registry (`src/blocks/__init__.py`), plus the supervised TCAV-style reference and the held-out readout / off-target probes.
4. Implement OOD loaders for R / Sketch / C(1–5) / ObjectNet with a fixed, matched evaluation protocol across shifts.
5. Wire `cfs_score`, `onmanifold_projection_residual`, and the `faithfulness(variant, cfg)` dispatcher so `run_experiments` and the EDA notebook share one scoring model; emit `cfs` + components + `offmanifold_residual` to `metrics_all.csv`.
6. Validate end-to-end on the offline synthetic planted-concept smoke task; confirm `onmanifold_steel` beats `naive/random` on synthetic-CFS.

Deliverable #2 — Pipeline & Implementation

A runnable pipeline: frozen CLIP → TopK SAE → concept directions → the four steering variants + supervised reference → CFS scoring, with all five OOD datasets loadable and the synthetic smoke path green.

Acceptance Check

`python -m src.run_experiments -smoke` writes `metrics_all.csv` with a row per variant including `cfs` and `offmanifold_residual`; `onmanifold_steel` shows lower off-manifold residual and higher synthetic-CFS than `naive/random`; the real CLIP + ImageNet-shift loaders import and return correctly-shaped batches on at least one shift level.

5 Milestone 3: Training & Ablation Experiments

Model / Steering method	Cache-or-cost	Primary metrics
Supervised TCAV-style (reference)	Label-expensive direction	CFS + components (the gold to approach)
<code>onmanifold_steel</code> (ours)	$P_M \Delta$, rank r	CFS, off-manifold residual, OOD slope
<code>clamp_steel</code>	Fixed-magnitude clamp, no projection	CFS, residual
<code>naive_steel</code>	Off-manifold $a + s d$	CFS, residual (main competitor)
<code>random_steel</code>	Random direction	CFS (null/sanity floor)

Implementation details. All methods are evaluated at matched steering strength on the same selected concept set; readout probes are held out from selection; the OOD ladder uses identical preprocessing per shift; CFS, its components, and the off-manifold residual are bootstrapped over concepts. The core experiment is the $r \times s$ design grid ($A3 \times$ steering-strength) feeding the headline CFS-vs-severity curve.

Error taxonomy

Error Type	Description	Expected Frequency
Off-Manifold Artifact	Naive edit leaves the real-image manifold; readout moves but the change is non-realistic / undecodable (a fake causal effect)	High for <code>naive_steel/clamp_steel</code> ; rare for <code>onmanifold_steel</code>
Specificity Leakage	Steering the target concept also moves off-target probes (entangled / polysemantic direction)	Medium; rises with steering strength
Polysemantic / Uninterpretable Feature	The SAE feature encodes several concepts (or none clean) $\Rightarrow$ no clean knob; filtered by A4	Medium pre-selection; $\sim 85\%$ of raw features
OOD Collapse	Faithful in-distribution but CFS collapses under shift (Sketch / C-high / ObjectNet) — the headline failure mode	Rises sharply past the shift knee
Magnitude Saturation	Beyond a strength threshold the readout flattens / clips; monotonicity breaks	Rises with steering strength

Ablations A1–A5

A1 — SAE type

TopK vs. L1 (vanilla) SAE: effect on clean concept magnitudes and CFS.

A2 — TopK k

Sparsity-level sweep: how k trades interpretability against faithfulness; CFS per k .

A3 — Projection rank r

The core knob: low r over-constrains (effect dies), high r lets the edit drift off-manifold; locate the CFS knee.

A4 — Concept-selection threshold

How strict the “well-defined concept” filter is, and the resulting reliable-concept fraction (the $\sim 10\text{--}15\%$ tail); CFS per threshold.

A5 — Backbone layer & token

Which ViT-B/16 layer, and patch vs. CLS tokens, for SAE training and steering; CFS per choice.

The core ablation is the **A3** $\times$ **steering-strength** 2-D grid (figures **fig5/fig3**).

Deliverable #3 — Experiments & Ablations

A complete **results/metrics_all.csv** covering all 5 steering methods across the full OOD ladder, the A1–A5 ablation grids (with the A3 $\times$ s core grid), and a populated error-taxonomy tally — all with bootstrap CIs over concepts.

Acceptance Check

Every (method, shift level) cell has a CFS with a confidence interval; **random_steer** sits at the floor and the supervised reference near the top (sanity); the A3 sweep shows a non-monotone CFS-vs-*r* curve with an identifiable knee; the strength sweep shows the saturation onset; each error type has a measured frequency.

6 Milestone 4: Analysis, Insights & Manuscript

Core visualizations (9).

- **fig1_cfs_ood_sweep** — **headline**: CFS vs. shift severity (clean $\rightarrow$ R $\rightarrow$ Sketch $\rightarrow$ C-1..5 $\rightarrow$ ObjectNet), on-manifold vs. naive; collapse knee marked, CI band.
- **fig2_faithfulness_pareto** — specificity vs. effect-size across all 5 methods; on-manifold on the frontier, random in the corner.
- **fig3_strength_sweep** — CFS vs. strength *s* with the saturation knee.
- **fig4_concept_reliability** — per-concept CFS distribution: the long unreliable mass and the $\sim 10\text{--}15\%$ reliable tail.
- **fig5_ood_heatmap** — shift type $\times$ method CFS grid; best cell boxed.
- **fig6_monotonicity_curve** — example readout vs. knob: smooth (on-manifold) vs. jagged (naive).
- **fig7_by_method_bar** — mean CFS by variant with bootstrap CIs; plus schematics **fig_overview** and **fig_method**.

Insight report. *RQ1*: whether on-manifold beats naive / random / clamp at matched strength on clean images and how close it gets to the supervised reference, with CIs. *RQ2*: decompose CFS into monotonicity / specificity / sufficiency, give the reliable-concept fraction against the $\sim 10\text{--}15\%$ claim, and locate the optimum over (*s*, *r*). *RQ3*: the CFS-vs-severity curve, the Δ CFS-per-shift slope, the collapse knee, and whether on-manifold degrades more gracefully than naive.

Use Case	Recommended Design	Reasoning
Trustworthy concept editing on clean data	<code>onmanifold_steer</code> , moderate r , pre-saturation s	Highest clean CFS at matched strength; low off-manifold residual
Auditing under domain shift	On-manifold + report CFS at each shift level	Tracks the collapse knee; flags where a “concept” stops being causal
Label-rich, accuracy-critical concept	Supervised TCAV-style reference	The expensive gold; use when labels exist and faithfulness is paramount
Quick unsupervised exploration	On-manifold over a strict A4-filtered concept set	Skip the $\sim 85\%$ polysemantic tail; spend effort on reliable knobs
Never	<code>naive_steer</code> / <code>random_steer</code> as evidence	High apparent effect, low real faithfulness — artifact or null

Manuscript outline (8 sections, IEEE conference). (1) Introduction & the open gap; (2) Background (vision SAEs, steering, faithfulness, robustness); (3) Method — the edit $a' = a + s(P_M\Delta)$ and the CFS definition; (4) Experimental setup; (5) Results — RQ1–RQ3 with the 9 figures; (6) Ablations A1–A5; (7) Discussion, limitations & the per-concept reliability story; (8) Conclusion. **Packaging:** reproducible config, `metrics_all.csv`, figure scripts, and data cards, all consistent with this roadmap and the brief.

Deliverable #4 — Analysis & Manuscript

The 9 figures rendered from real M3 results; an insight report answering RQ1–RQ3; the practitioner decision table; and a complete IEEE-conference manuscript draft.

Acceptance Check

Each RQ has a one-paragraph, evidence-backed answer with CIs; every figure caption states the measured (not illustrative) numbers; the decision table’s recommendations follow from the results; the manuscript compiles and matches this roadmap and the brief on title, RQs, baselines, benchmarks, and variant names.

7 One-Page Checklist

Weeks 1–2 — Literature & Design

- ☐ Read & annotate all 12 papers; draft the ~ 24 -entry bibliography.
- ☐ Fix the CFS definition and the on-manifold edit $a' = a + s(P_M\Delta)$.
- ☐ Specify the 4 variants + supervised reference; matched-strength protocol.
- ☐ Deliverable #1 + Acceptance Check signed off.

☐ Four steering variants + readout / off-target probes implemented.

☐ OOD loaders (R / Sketch / C / ObjectNet) and CFS dispatcher wired.

Weeks 3–4 — Data & Implementation

- ☐ CLIP ViT-B/16 activations cached; TopK SAE trained; U_r estimated.

☐ Synthetic smoke green; Deliverable #2 + Acceptance Check.

Weeks 5–6 — Experiments & Ablations

- ☐ CFS for all 5 methods across the full OOD ladder, bootstrapped.
- ☐ Ablations A1–A5, including the $A3 \times s$ core grid.
- ☐ Error taxonomy tallied; `metrics_all.csv` complete.
- ☐ Deliverable #3 + Acceptance Check.

Weeks 7–8 — Analysis & Manuscript

- ☐ All 9 figures rendered from real results.
- ☐ Insight report answers RQ1–RQ3 with CIs; decision table filled.
- ☐ IEEE-conference manuscript drafted; consistent with the brief.
- ☐ Deliverable #4 + Acceptance Check; repository packaged.

8 Quick Reference Links

- Cunningham et al., *SAEs Find Interpretable Features* — [2309.08600](#)
- Gao et al., *Scaling & Evaluating SAEs (TopK)* — [2406.04093](#)
- Turner et al., *Activation Addition* — [2308.10248](#)
- Zou et al., *Representation Engineering* — [2310.01405](#)
- Kim et al., *TCAV* — [1711.11279](#)
- Radford et al., *CLIP* — [2103.00020](#)
- Dosovitskiy et al., *ViT* — [2010.11929](#)
- Hendrycks et al., *ImageNet-R* — [2006.16241](#)
- Hendrycks & Dietterich, *ImageNet-C* — [1903.12261](#)
- Wang et al., *ImageNet-Sketch* — [1905.13549](#)
- Makelov et al., *Principled SAE Eval* — [2405.08366](#)
- Marks et al., *Sparse Feature Circuits* — [2403.19647](#)